

Rajan Kumar Singh

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EDUCATION

- **VIT Bhopal University** Bhopal, India
Bachelor of Technology in Computer Science And Engineering; CGPA: (8.14/10.0) *Aug. 2022 – July. 2026*

PROJECTS

- **Cloud-Crafters** May 2025 - Ongoing
Full Stack AI-Powered platform for mapping and generating code
 - **Service-mapping:** Developed a web platform to map cloud services across AWS, Azure and GCP, enabling cross-cloud migration knowledge with configuration equivalence.
 - **Terraform Code:** Building an AI-Powered Terraform generator using OpenAI API to automate infrastructure-as-code creation.
 - **Service Creation Wizard:** Designed a step-by-step Service Creation Wizard for deploying services including prerequisite checks, config suggestions, and real-time validation.
 - **Databases:** Implemented PostgreSQL schema and Prisma ORM to manage users deployment templates, session states and infrastructure history.
 - **Authentication:** Integrated NextAuth.js for authentication and AI API tokens using CryptoJS encryption, ensuring safe user configuration and access
- **SnapCache** Jan 2025 - March 2025
Redis-Style In-Memory Key-Value Store Server
 - **Server:** Developed a Redis-like in-memory key-value store in c++ using low-level TCP sockets..
 - **Parser:** Developed a custom commands(GET, SET, EXPIRE) with event-driven architecture .
 - **Data-structure:** Used unordered_map and TTL logic with background thread for automatic key expiry and cleanup.
 - **Pub/Sub:** Implemented basic Pub/Sub mechanism , binary serialization.
 - **Parser:** Gained Hands-on experience with multithreading, memory management , and low-level network programming in C++.

TECHNICAL SKILLS

- **Programming Languages Frameworks:** C++, Typescript, Next.js 14, React.js, Node.js
- **Backend Database:** PostgreSQL, TCP/IP, Socket Programming, Zod, Concurrency.
- **Cloud Infrastructure:** Terraform, AWS, AI Integration.
- **Security Devtools:** JWT, bcryptjs, Axios, GIt, Github, VSCode , Neovim.

EXTRA-CURRICULAR

- **Documentation Writing:** Wrote detailed project documentation while building Cloud Service Mappers and SnapCache focusing on making workflow easier and faster.
- **Cloud Platform Certification :** Certification from educational platform like ethnus on AWS cloud practitioner and MERN stack development.
- **Open-Source:** Explored Open source tools like shadcn/ui and Terraform modules.
- **Community Engagement:** Active in online tech communities (Github, Stack Overflow, Reddit).